Project Title: ABC (Amazing Business Computing)

Problem Statement

Based on a survey that we conducted, 3 out of 5 student entrepreneurs experience difficulties in tracking their own business transactions and managing their own finances. Small businesses may handle an estimated 10–20 transactions daily, equivalent to approximately 300–600 transactions monthly. Manually recording item prices, tracking purchases, and comparing sales revenue with available capital can be time-consuming and prone to errors, particularly when handling multiple items. Business owners may use calculators, notebooks, or spreadsheets to record transactions and compute their earnings, but these methods still require manual input and calculations.

The ABC (Amazing Business Computing) project aims to address these challenges by developing a menu-based Python application that records item names and prices, allows users to set their available capital or budget, stores transaction records, and automatically computes potential profit or loss. The program will accumulate the entered item prices as total revenue and compare this amount with the user's budget to determine the difference. If the total revenue exceeds the budget, the program reports a profit; otherwise, it reports a loss or no profit, depending on the result. By automating these calculations and organizing records, the application aims to reduce arithmetic errors and save users time when reviewing their transactions.

Project Objectives

1. To develop a menu-driven Python application that automates basic financial tracking for small businesses and student entrepreneurs using loops, conditional statements, functions, lists, and input validation.
2. To implement a data-entry system that allows users to record item names and prices, set their available capital or budget, and review previously stored records through an organized menu interface.
3. To automate the accumulation of entered item prices and comparison with the user's budget or capital, displaying the resulting financial difference to reduce repetitive manual calculations.
4. To improve the efficiency and accuracy of financial recordkeeping by organizing transaction data and minimizing errors associated with manual calculations, particularly when handling multiple entries.
5. To demonstrate the potential usefulness of automation in managing an estimated 300–900 transactions per month, based on an illustrative estimate of 10–30 transactions per day, and how automated recording and calculations can simplify financial tracking.